Produce constructor standings

```
dbutils.widgets.text("p_file_date", "2021-03-28")
v_file_date = dbutils.widgets.get("p_file_date")
```

```
%run "../includes/configuration"
```

```
%run "../includes/common_functions"
```

```
# race_results_df = spark.read.parquet(f"{presentation_folder_path}/race_results")
```

```
race_results_df = spark.read.format("delta").load(f"{presentation_folder_path}/race_results") \
.filter(f"file_date = '{v_file_date}'")
```

```
race_year_list = df_column_to_list(race_results_df, 'race_year')
```

```
from pyspark.sql.functions import col

race_results_df = spark.read.format("delta").load(f"{presentation_folder_path}/race_results") \
.filter(col("race_year").isin(race_year_list))
```

```
from pyspark.sql.functions import sum, when, count, col

constructor_standings_df = race_results_df \
.groupBy("race_year", "team") \
.agg(sum("points").alias("total_points"),
     count(when(col("position") == 1, True)).alias("wins"))
```

```
# display(constructor_standings_df.filter("race_year = 2020"))
```

```
from pyspark.sql.window import Window
from pyspark.sql.functions import desc, rank, asc

constructor_rank_spec = Window.partitionBy("race_year").orderBy(desc("total_points"), desc("wins"))
final_df = constructor_standings_df.withColumn("rank", rank().over(constructor_rank_spec))
```

```
# display(final_df.filter("race_year = 2020"))
```

```
# final_df.write.mode("overwrite").format("parquet").saveAsTable("f1_presentation.constructor_standings")
```

```
merge_condition = "tgt.team = src.team AND tgt.race_year = src.race_year"
merge_delta_data(final_df, 'f1_presentation', 'constructor_standings', presentation_folder_path, merge_condition, 'race_year')
```

```
# %sql
# select * from f1_presentation.constructor_standings;
```

Table

	race_year	team	total_points	wins	rank
64	1959	Cooper-Borgward	0	0	8
65	1959	Cooper-OSCA	0	0	8
66	1959	Aston Martin	0	0	8
67	1959	Cooper-Maserati	0	0	8
68	1959	Vanwall	0	0	8
69	1959	JBW	0	0	8
70	1989	McLaren	141	10	1
71	1989	Williams	77	2	2
72	1989	Ferrari	59	3	3
73	1989	Benetton	39	1	4
74	1989	Tyrrell	16	0	5
75	1989	Team Lotus	15	0	6
76	1989	Arrows	13	0	7
77	1989	Brabham	8	0	8
78	1989	Dallara	8	0	8
79	1989	Onyx	6	0	10

1,081 rows

Table

	race_year	count(1)
1	2021	10
2	2020	10
3	2019	10
4	2018	10
5	2017	10
6	2016	11
7	2015	10
8	2014	11
9	2013	11
10	2012	12
11	2011	12
12	2010	12
13	2009	10
14	2008	11

15	2007	11

72 rows